

Extreme Rainfall Practical

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1 Introduction

In general, rain is produced when a parcel of air rises, cools, and releases its moisture. Warmer air can contain more moisture than colder air, so as warmer air cools, water vapour condenses onto particles such as dust and aerosols (MetOffice, 2019). These particles are called cloud condensation nuclei (CCN). The rate of formation of these droplets depends on the number of CCN and the (super)saturation of the air parcel (PSU, 2023). Once a droplet forms it grows through both additional condensation and coalescence (colliding and combining with other droplets) (University of Reading, 2011) Once a water droplet is large enough it will fall out the cloud as rain.

The MetOffice categorise rain into three categories depending on its driving force: convective, frontal, and orographic (MetOffice, 2019).

Convective is the most simple. The ground is warmed by the incoming solar radiation, causing surface air to warm and rise as it is now less dense than its surroundings. The warm air then cools and water droplets condense out. This type of rain can produce patchy showers as the ground is heated up unevenly due to topography differences. In unstable atmospheric conditions (dry adiabatic lapse rate $<$ environmental lapse rate), convective rain can be associated with storms - if the package of air continues being warmer than its surroundings, it will continue to rise, releasing more water.

Frontal rain causes persistent long term rain. When a warm body of air and a cold body collide at a weather front, the warm air is pushed above. It then cools and releases water, causing rain. Due to the size of fronts, cloudy conditions and rain can last for long periods of time (up to a day).

Orographic rain is similar to frontal rain, however, hills are the upwards driving factor rather than a body of cold air. Wind can force air up and over a hill and the package of air will cool with the altitude change.

Rainfall during summer is frequently due to convective rain. The warmer summer months will have higher incoming solar radiation, leading to increased ground warming. Conversely, winter rain is driven by larger scale weather systems, leading to more frontal rain bringing long-term overcast drizzly skies.

The data for this report are taken from the MetOffice's Hadley Centre observations datasets (Alexander and Jones, 2001). This dataset contains daily rainfall measurements for 9 different areas across the UK, grouped into three major area: England and Wales, Scotland, and Northern Ireland. The dataset starts in January 1931 and goes up to current day (March 2023). This report focuses on the Scottish rainfall data, ignoring 2023 as the year is incomplete.

2 Methods and Results

2.1 Monthly Variations

The data from the MetOffice (Alexander and Jones, 2001) were grouped by month and the maximum daily rainfall per month found. This data was then grouped through years, taking a total mean of the maxima of each given month. This is plotted in figure 1, with the black line showing the mean of the monthly maxima. Note that the blue line shows the minimum of the maximum values per month, rather than the actual minimum values. A clear seasonal trend can be seen from the black line, with the average wettest days being towards the end of the year, and the driest from March to June. The monthly maxima are shown to follow the same trend with the wettest day in the dataset being recorded on the 3rd October 2020 with 52.8mm. The driest of the wettest days of each month was the 9th February 1932. The data can also be represented more neatly as a set of violin plots as shown in figure 2. A violin plot also shows the distribution of data points between the minima and maxima. January to May have almost normally distributed data, with the majority at the mean, then spreading out toward the minimum and maximum points. Whereas, June to November seem bottom heavy, with less data points reaching close to the maxima and larger groupings below the mean.

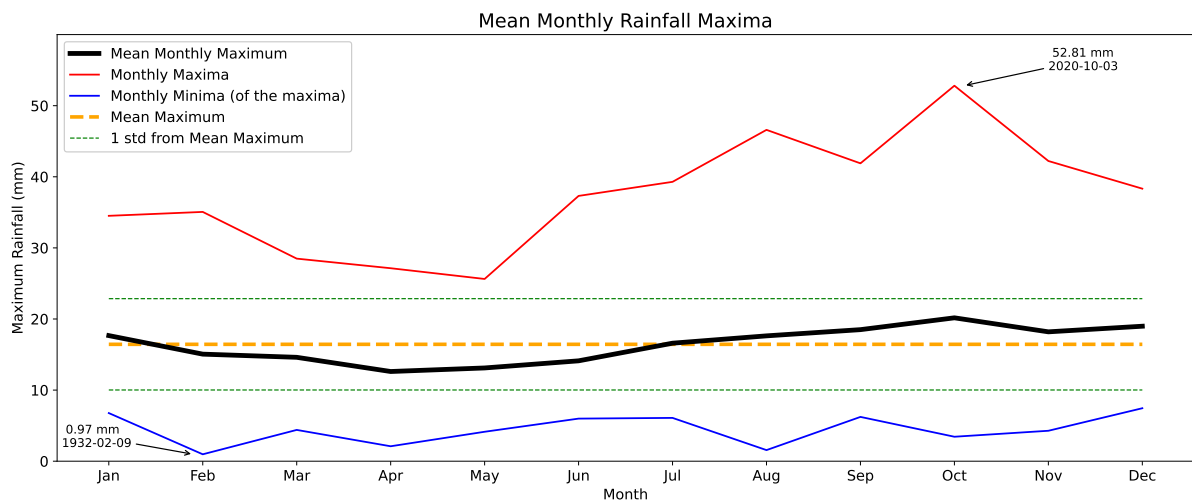


Figure 1: A graph showing monthly variation in maximum daily rainfall.

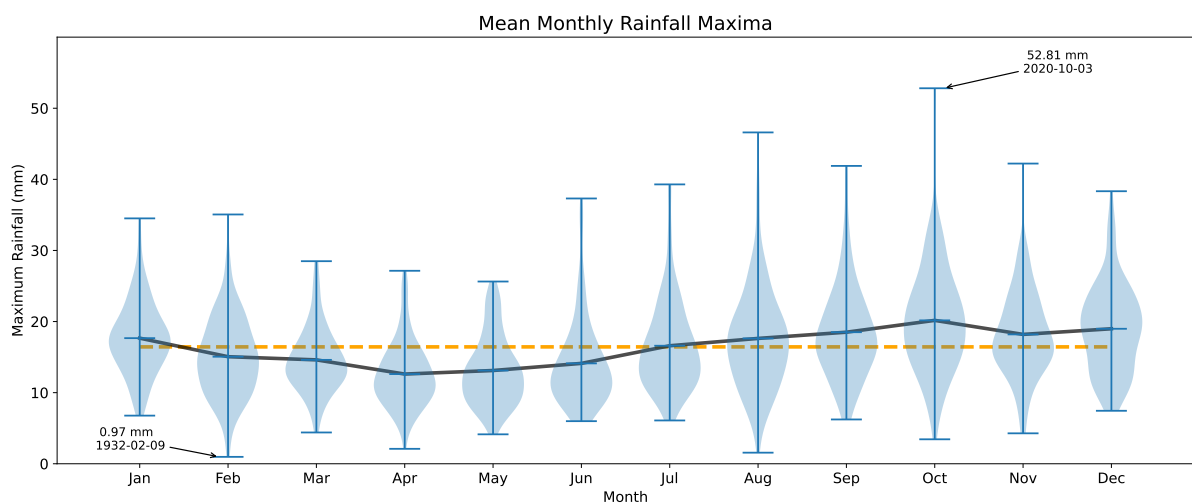


Figure 2: A grouped violin plot showing monthly variation in maximum daily rainfall.

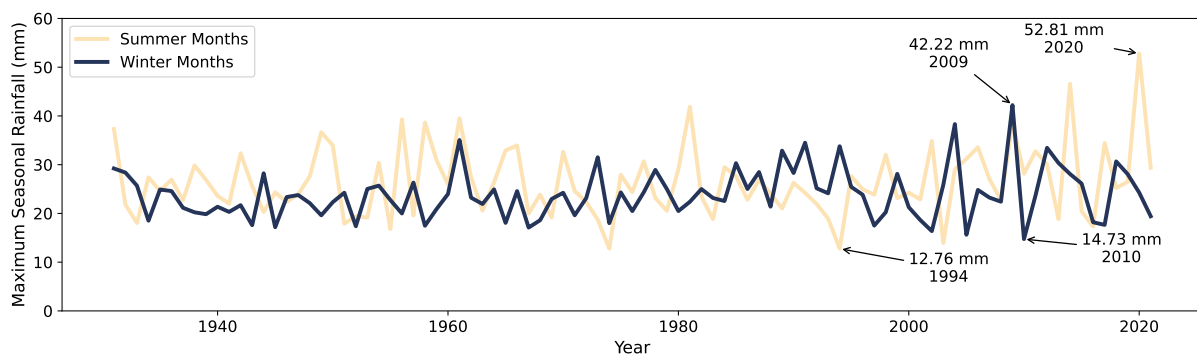


Figure 3: A timeseries showing the maximum daily rainfall across each season from 1931-2021.

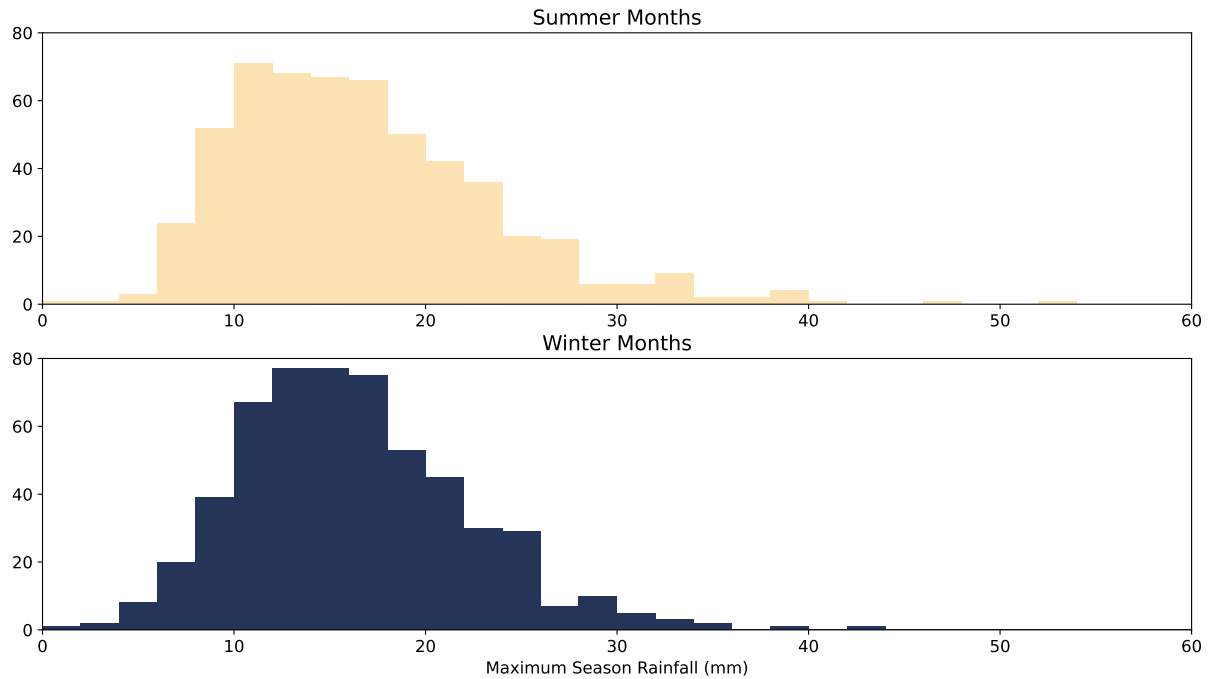


Figure 4: Two histograms showing the frequency of maximum daily rainfall during winter and summer months.

2.2 Seasonal Analysis

To Further examine the seasonal variations, the dataset was grouped into summer (ranging from May-October) and winter (November-April) months. Care was taken to correctly group the winter months of the same year with each-other, i.e. January-April 2012 and November-December 2011 are grouped under the label winter 2011. This meant that the starting (1931) and end (2022) years of the dataset's winters are incomplete so they were discarded. Summer months of those years were also discarded to keep the data equal lengths. The maximum overall values from each season were taken and plotted in a timeseries, shown in figure 3. More extremes tend happen in the more recent years (<30 years ago), though the trend is not clear from this graph alone.

The seasonal data can also be examined in histograms. Figure 4 shows the frequency of monthly daily rainfall maxima separated by summer and winter months. Both summer and winter have 10-20mm as their most common maxima, however, summer months have higher extremes. The value of ~53mm stands out far from the rest of the data. Winter's most intense extremes have peak rainfall around the 40mm mark rather than the 50mm of summer. Additionally, summer has more values between 30-40mm than winter. To understand whether extreme events have been increasing or decreasing over the past 90 years, the data can be separated into year groups - one from 1931-1976 and another from 1976-2021. This is shown in figure 5. Pre-1976 summer does not have any days over 40mm, however post-1976 has three separate events. Similarly, winter has its only >40mm day after 1976.

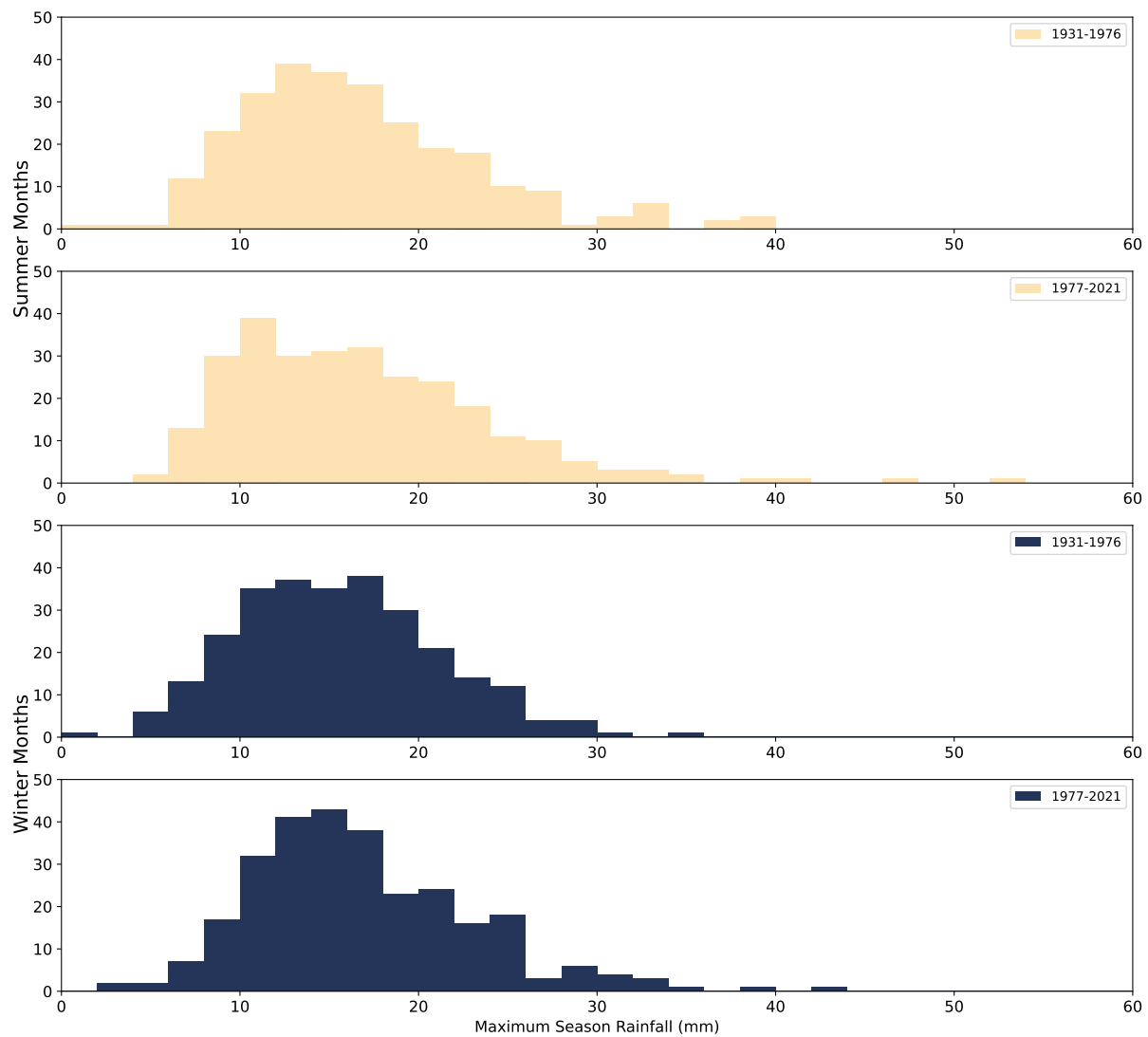


Figure 5: Four histograms showing the frequency of maximum daily rainfall during winter and summer months, seperated across 1976.

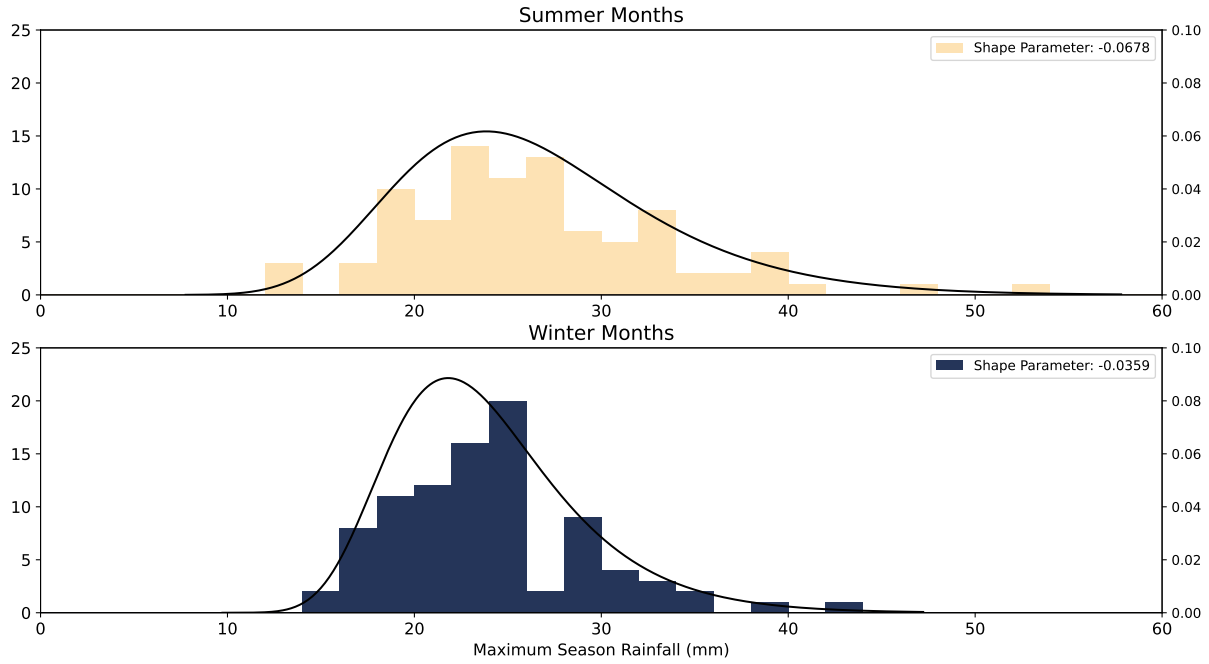


Figure 6: Two histograms showing the frequency of seasonal maxima of daily rainfall during winter and summer months. The best fit GEV from scipy is overlain. The shape parameter of both distributions is shown in the top right of each graph.

2.3 GEV Fitting

The next step when examining extremes is to fit the data to the Generalised Extreme Value (GEV) distribution.

Important note: the Python package, scipy, uses a slightly different equation, shown in equation 1 (Scipy, 2023), to that which was learnt in class, shown in equation 2.

$$f_{\text{PDF}}(x, c) = e^{-(1-cx)^{\frac{1}{c}}} (1 - cx)^{\frac{1}{c}-1} \quad (\text{Scipy, 2023}) \quad (1)$$

$$F_{\text{CDF}}(y \mid \mu, \sigma, \xi) = e^{-1 + \xi \frac{y - \mu}{\sigma} - \frac{1}{\xi}} \quad (2)$$

The practical difference is that scipy's shape parameter (c) is equal and opposite to the lectures' ξ . For example, a Weibull distribution may have a shape parameter of 0.25 in scipy, but -0.25 in class. For the purpose of this report, the class convention will be used, so the results may be different when going to reproduce the shape parameter values in Python. The scipy shape parameter sign convention is discussed further in the GitHub commits (ev-br, 2014).

The maximum seasonal values (as plotted in figure 3) were analysed using scipy's GEV fit method, which uses a negative log likelihood function to fit the data to a GEV distribution. The resulting GEV probability density function (PDF) is shown in figure 6, along with the relevant histograms. Both GEV PDFs fit the histograms well, with summer's spreading over a wider range. Both returned positive shape parameters from scipy, which correspond to negative shape parameters in the lectures' convention. This makes both distributions bounded Frechet distributions. Summer's shape parameter is more negative, implying it is a more bounded distribution, despite summer having more extreme values.

To find the intensity of an event from a specific return period, the probability of the event in a given time frame first needs to be found. For a 10 and 50 year return period, the probability of each in a year is $\frac{1}{10} = 0.1$ and $\frac{1}{50} = 0.02$ respectively. These probabilities can then be mapped onto the CDF and a value for maximum rainfall found. This process is shown in figure 7. Summer shows higher intensity events than winter. In order to reach the peak of $\sim 53\text{mm}$ seen on 3/10/20, a value of ~ 0.9975 is obtained from the summer CDF - corresponding to a return period of ~ 400 years! Similar analyses were carried out on two more groups of data. One filtering from 1932-1991 and the other from 1962-2021. The results are shown in table 1.

Summer has more extreme values than winter for all date ranges. Both seasons show decreasing intensity when only using older data, and higher intensity when using more recent. Summer's shape parameter appears more short tailed with older data, however, winter shows the most negative shape parameter with the newest data.

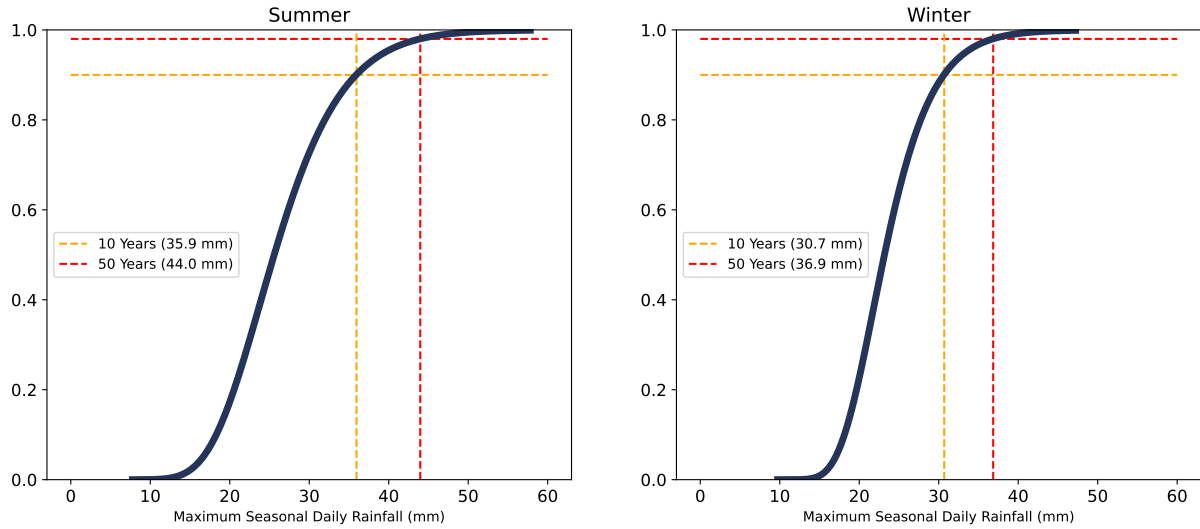


Figure 7: The summer and winter seasonal maximum data fitted to a GEV CDF. Lines are drawn on to show the intensity of 10 and 50 year return period events.

Season	Data Range	Shape Parameter	10-Year Event (mm)	50-Year Event (mm)
Summer	All Data	-0.0678	35.9	44.0
	1932-1991	-0.1000	34.1	40.7
	1962-2021	-0.0686	36.1	44.5
Winter	All Data	-0.0359	30.7	36.9
	1932-1991	-0.0515	29.1	34.0
	1962-2021	-0.0612	31.8	38.1

Table 1: A table showing the event intensities for 10 and 50 year return period events with varying date ranges.

2.4 Droughts

Measuring the opposite of intense rainfall (droughts) can be difficult. One day with no rain does not indicate a drought, and the concept of droughts itself has over 150 definitions (NIDIS, 2023). Another metric or quantifier needs to be defined. Number of days below a specific threshold of rain could be relevant - a longer period with no rain leads to a more extreme impact.

An arbitrary threshold for ‘enough rain to end drought’ was chosen as one fifth of the overall mean daily rainfall. The number of consecutive days below that threshold was calculated in order to find the length of each drought. A histogram displaying this data is shown in figure 8a. The graph has been cut off at a threshold of 10+ days so that the extremes are more visible.

The maximum yearly value was then selected to find the longest, most extreme, drought of each year. This was then plotted as both a histogram and timeseries in figures 8b and 8c. The histogram resembled the shape of a GEV distribution so GEV PDF was fitted and overlaid using the same method described in section 2.3. The shape parameter was negative, indicating a bounded, Frechet distribution. The same analysis as section 2.3 can be carried out and a table of return values produced in table 2. This shows the opposite trends to the rainfall extremes, with 10 and 50 year events being less extreme in more recent data than older data.. A positive shape parameter was found for older events, indicating unbound, whereas more recent data is bound.

Data Range	Shape Parameter	10-Year Event (days)	50-Year Event (days)
All Data	-0.0245	18.5	23.4
1932-1991	0.1368	19.4	27.0
1962-2021	-0.1782	17.1	20.3

Table 2: A table showing the drought length times (in days) for 10 and 50 year return period events with varying date ranges.

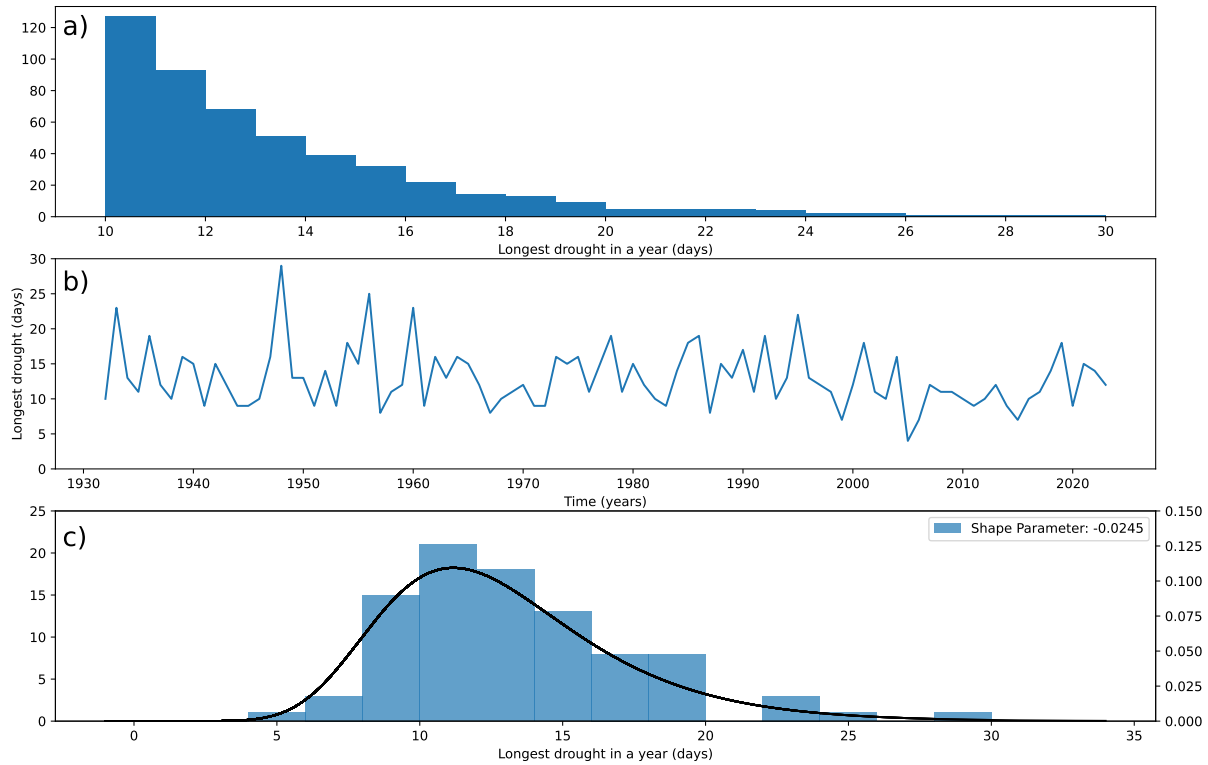


Figure 8: Three graphs showing the distribution of length of droughts. a) shows the frequency of droughts above 10 days long since 1931. b) shows a timeseries of the longest drought per year. c) shows the frequency distribution of the longest droughts per year shown in b).

3 Discussion

3.1 Rainfall

Figures 1 and 2, found that more intense extremes happen during the summer and autumn months, with October having the highest average maximum rainfall and the highest maximum rainfall in a 24 hour window overall. February to June showed below average sized extremes, with February having the smallest monthly maximum on record. These results are corroborated by the findings in the summer/winter analysis - with summer having higher frequency and higher magnitude extremes. As rainfall extremes in summer are fueled mainly by convective rain, this could be expected, as convective storms bring large, short-term rains, as opposed to winters long-term frontal rain. Despite this difference, winter can still bring intense extremes with a 50 year event only being ~6mm less.

There was significant difference between more recent and older data. An important statistic, as noted in section 2.2, is that pre 1976, there had never been a day with >40mm of rain for both summer and winter. The return periods also show increases in intensity, with a 50 year event bringing ~4mm more rainfall in more recent years for both summer and winter. This information shows that events are becoming more extreme with time. Additionally, the frequency of events is increasing, with strong increases in the 30-40mm extremes for both seasons.

An interesting note is the shape parameters of the winter and summer months. Rainfall is generally expected to be an unbounded distribution as there is no limit on how much moist air can be advected and produce rain. However, both seasons had negative tails, implying bounded processes - meaning there is a maximum limit on how extreme a rainfall event could be. One possibility could be that the data was not 'extreme enough'. Perhaps blocking the data into 5 year groups and finding the maximum could have fit an unbounded distribution. Alternatively, there might not have been enough data to show the fat tail expected from rainfall, and Scotland is overdue some heavy storms. The shape parameters have very small magnitudes meaning that they perhaps resemble a Gumbel distribution more so than a Frechet, but the consistent negative across all 6 values in table 1 support a bounded interpretation of the data. The same data was tested using another Python package for fitting GEV distributions called `lmoments3` (pascalbo, 2014), which also found all negative values, further supporting the bounded results.

3.2 Drought

Section 2.2 found that droughts had been decreasing in length with time. It also found that drought length could be fitted to a GEV distribution, where more recent 50 year events were less intense. These findings are supported by a 2017 report from the UK MetOffice (MetOffice, 2017), that found dry spells had not been increasing in length with time. However, the MetOffice acknowledges that drought length is not a good metric for measuring the impact of a drought (MetOffice, 2023). They forecast that the total number of dry days will increase, with wet days becoming more intense (as shown in section 2.2), but not necessarily with long dry spells. More inconsistent water leads to drying out of the ground and more difficult water supply challenges. Figure 8 struggles to identify significant droughts in UK history. The 1976 drought is not a significant blip on figure 8b, supporting that this method is a poor indicator of drought intensity and impact.

4 Impacts

The most intense rain event in the data is the 3rd of October 2020, as noted in figures 1 and 2. A MetOffice blog post (MetOffice, 2020b) confirms this, stating that it was the wettest day on record and ‘could have more than filled Loch Ness’. Figure 9 shows the distribution of the rainfall across the UK with east Scotland having the highest rainfall. The weather system responsible was an extratropical cyclone called storm Alex (MetOffice, 2020a). A low pressure centre, shown in figure 10, made its way southeast towards the UK and Europe. Large frontal systems brought extensive rainfall to the UK, causing temporary power cuts in multiple UK locations. Despite a relatively dry September causing dry ground, flooding impacts were limited in the UK (MetOffice, 2020a). Most devastating impacts were seen in France, with at least 5 fatalities from flooding and landslides (BBC News, 2020) and thousands of homes losing power. Areas of Italy were also severely impacted with as much as 630mm of rain falling in some areas and wind speeds up to 112mph.

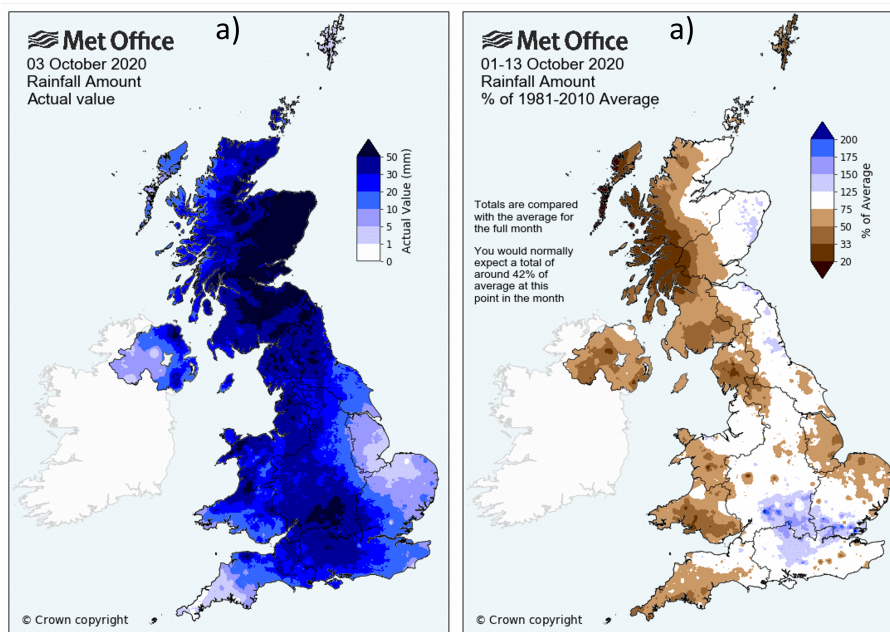


Figure 9: Two maps showing the rainfall distribution across the UK on the 3rd of October 2020 (MetOffice, 2020b). a) shows the rainfall in mm on the day. central and southern England and east Scotland show the highest values. b) shows the cumulative rainfall in the first third of the month as a percentage of the total monthly rainfall. Similar areas show to a) show values above the expected 42%.

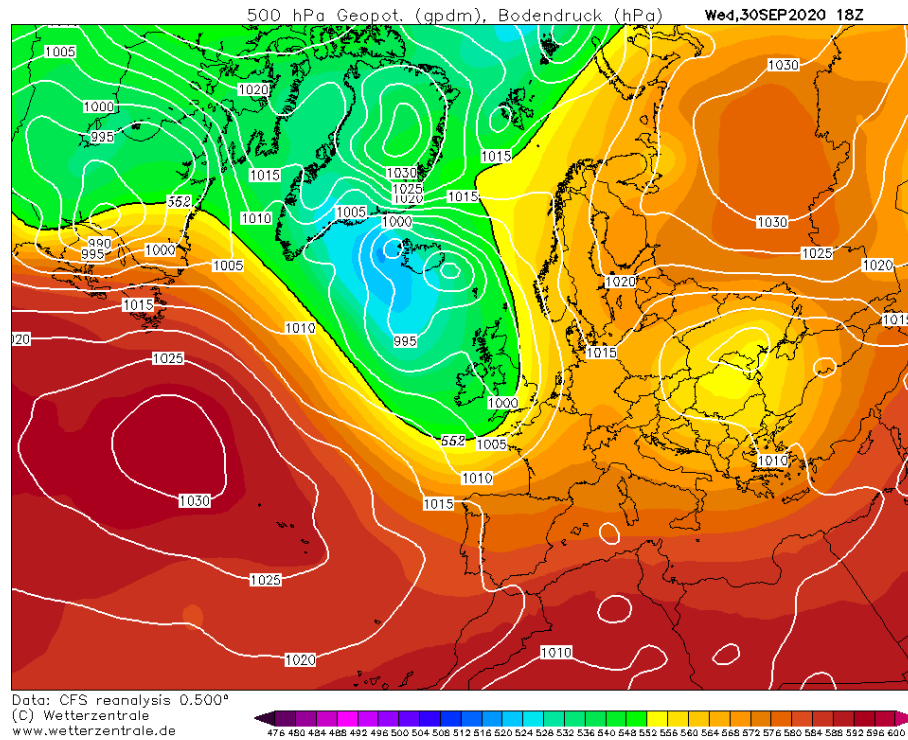


Figure 10: A pressure map showing the starting point of the low pressure system that led to storm Alex. The date is the 30th September - 3 days before the 3rd October. (Wetterzentrale, 2020).

5 Conclusion

Sections 2.1, and 2.2 showed that the warmer summer months had generally more extreme rainfall than the summer months, potentially due to summer rain being driven by convective storms. That being said, the largest rainfall day on record, was in October (categorised as a summer month), and the extreme rainfall was due to frontal activity from an extratropical storm rather than a convective storm. Fitting to a GEV distribution showed that extremes had been getting more intense in more recent years, although the GEV fit was unexpectedly bounded. Drought analysis showed good promise, agreeing with the MetOffice analysis that dry spells are getting shorter. However, it must be acknowledged that dry spells alone are a poor indicator of drought severity and impact.

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